

**DAYANANDA SAGAR COLLEGE OF ENGINEERING**

(An Autonomous Institute Affiliated to VTU, Belagavi, Accredited by NAAC with 'A' Grade)

Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560111

Department of Robotics and Artificial Intelligence

Continuous Internal Assessment Test- 1

Course: Image Processing & Its Application

Maximum marks: 50

Course Code: 22RI63

Date: 20/02/2026

Semester: 6th

Duration: 90 Min

Question No.	1	2	3	4	5	6	7	8
Course Outcome	1	2	1	1	2	2	2	2
Program Outcome	1,2	1,2	1,2	1,2	1,2	1,2	1,2	1,2
Blooms Level	2	3	2	2	3	3	3	3

Answer five full questions, questions 1 and 2 are compulsory all questions carry equal marks

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1	Illustrate the fundamental steps in image processing with a block diagram	10
2	Identify the types of image segmentation techniques and explain its importance	10
3	Elucidate image sensing and image acquisition process	10
OR		
4	Explain the components of image processing system	10
5	Illustrate the working of smoothening frequency domain filters	10
OR		
6	Illustrate the working of sharpening frequency domain filters	10
7	Apply image filtering techniques on different noise models	10
OR		
8	Apply morphological operations and explain its importance in image processing	10



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Department of Robotics and Artificial Intelligence

Continuous Internal Assessment Test- 3

Course: **Image Processing and its Application**

Maximum marks: **50**

Course Code: **22RI63**

Date: **5/5/2026**

Semester: **6TH**

Duration: **90 Min**

Question No.	1	2	3	4	5	6	7	8
Course Outcome	CO4	CO5	CO4	CO4	CO5	CO5	CO5	CO5
Program Outcome	1,2,3,5,12	1,2,3,5,12	1,2,3,5,12	1,2,3,5,12	1,2,3,5,12	1,2,3,5,12	1,2,3,5,12	1,2,3,5,12
Blooms Level	L4	L4	L4	L4	L4	L4	L4	L4
Blooms Level Weightage (%)	L1	L2	L3	L4	L5	L6	Total BL Weightage (%)	
				100			100	
Course Outcome Weightage (%)	CO1	CO2	CO3	CO4	CO5	CO6	Total CO Weightage (%)	
				40	60		100	

Answer five full questions, questions 1 and 2 are compulsory all questions carry 10 marks

		Marks
1	Analyze the architecture of AlexNet	CO4
2	Examine the working of YOLO algorithm	CO5
3	Analyze the architecture of VGG 16	CO4
	(OR)	
4	Inspect the two stage object detection method	CO4
5	Examine how skip connections affect neural network with ResNet architecture	CO5
	(OR)	
6	Analyze inception module and dimensionality reduction on GoogLeNet architecture	CO5
7	Examine Frame-to-frame motion estimation in visual odometry	CO5
	(OR)	
8	Inspect working and types of Simultaneous Localization and Mapping (SLAM)	CO5